



Economics

National Accounting

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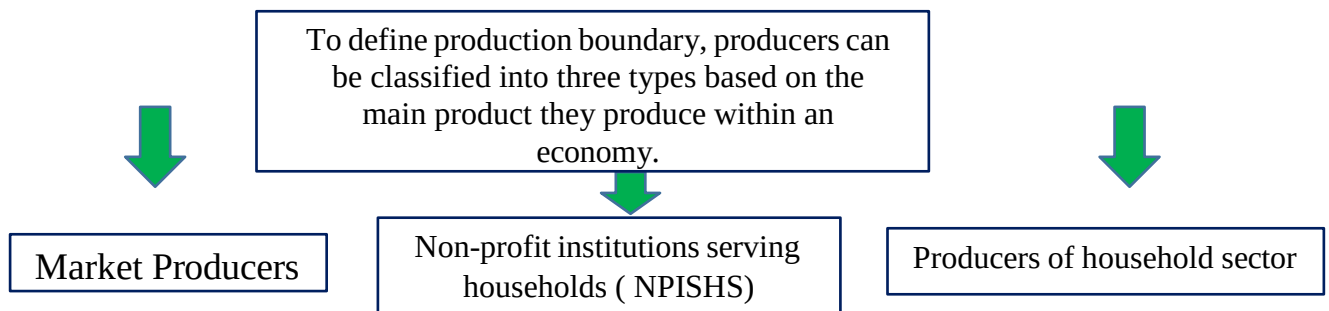
-National Accounting-

National accounting is the method of recording all types of financial transactions that take place in an economic area during a given financial year.

- It is essential to study all types of economic activities that take place in the economy. Thus, all human behavior has some economic value. But it really needs to be clarified which of these economic behaviors are effective economic activities and which are not.

Production boundary

The concept of Production boundary is used to define economically productive activities and non-economic productive activities separately.



○ production activities that should be included in the production boundary.

- All goods and services sold at the market are included in the production boundary.
- All goods and services that are produced for institutional units other than for the institution produced.
- All goods that are retained by their producers with themselves having produced them for their own final consumption or for purchasing of capital.
- The own-account production of housing services by owner occupiers.
- Domestic and personal services performed by paid domestic staff
- Producing of agricultural crops for own consumption, storing of those, gathering of fruits and obtaining of foods from forests, chopping and gathering of wood, hunting and catching fish.

- Although being a natural resource, if output is generated with the intervention of man, then it is included in the production boundary.

Example:- Growing of fish in a fish tank

○ production activities that should not be included in the production boundary.

- The preparation of foods for members of a household.
- Looking after the children.
- The care of old and infirm people of a household.
- Activities such as decorating and cleaning of a house.
- Growth of fish without intervention of man
- Unpaid household work.
- Freely obtained natural resources (Land, water, air etc.).
- Natural resources grown without the intervention of man (such as forests).
- Education and leisure.
- Change in value of resources due to natural growth or change in prices.

➤ Indicate whether the following economic activities are included in the production boundary or not.

- 1- 100 Kg of home grown chillies
- 2- A modern car for sale
- 3- Chairs made by a carpenter for his own use
- 4- Setting up a flower fence in the house
- 5- A newspaper bought from a shop

Non observed economic activities should also include in estimating the value of gross domestic product

○ There are two types of non-observed economic activities.

- Hidden economic activities

Hidden economic activities are those legal economic activities that are not captured in formal statistical enquiries.

Reasons affecting to hide these economic activities purposely.

- To avoid payment of tax

- To avoid the payment of social security contributions
- To avoid having to meet certain legal standards

Example:- Payment of minimum wages, laws relating to maximum work hours.

- Illegal economic activities

In this way due to absence of keeping records avoidance of laws and regulations and due to absence of information, data relating to hidden economic activities may not be obtained.

Two forms of illegal economic activities.

The production and distribution of goods and services forbidden by law.

Example:- Alcoholic , drugs, prostitution, transportation of goods prohibited

Some legal goods and services produced and distributed by unauthorized people.

Example:- Unlicensed medical practitioners

○ Nonproductive transactions excluded in national income accounting.

- Nonproductive market transactions
- Transfer payments
- Transactions relating to financial papers
- Transactions relating to second hand products

➤ Indicate whether the following economic activities are included in the Gross domestic production or not.

1. Charging money and committing murders
2. Buying a cell phone from a retail store
3. Keeping a sack of paddy for home use
4. Buying and selling Treasury bills

5. Government grant of Rs. 5000 / - to housing units.
6. A ticket purchased for a cricket match
7. A shirt bought the day before watching a cricket match.
8. Monthly salary of a woman working in the house
9. Payment for online classes teacher
10. The preparation of foods for members of a household
11. Growing of fish in a fish tank
12. Weaving cloth, dress making, making furniture
13. The supply of water by government
14. Growth of fish without intervention of man

The calculation of National accounting can be done through three approaches.

Product approach

Business sector produces goods and services by combining factors of production obtained from the household sector and it is **the product approach** relevant to national income accounting.

Income approach

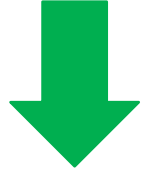
When the business sector purchases factors of production from the household sector there will be a flow of factor income flow to the household sector and it is **the income approach** in national income accounting.

Expenditure approach

Expenditures incurred by household sector to purchase goods and services produced by business sector is **the expenditure approach** in national income accounting.

Product approach

product approach is mainly the calculation of the Gross Domestic



Product.

Gross Domestic Product means the value of all final goods and services produced within a particular period of time (Normally within one year), within the geographical

when estimating the value of the Gross Domestic Product, the product's value can be overestimated by adding the value of same good at several times.

EX-

Firms	Product	Market price
1	Cotton	500
2	Threads	650
3	Fabric	900
4	clothes	1250



With this situation The same value is recorded several times through the accounting of the values of all the above institutions. That is, one product here includes the value of the other product.

For example, the production of yarn may contain cotton.

This addition of the value of the same good at several times is called multiple counting error.

Therefore, to estimate the value of the national output by eliminating the multiple counting error two approaches are being used.

1. Final product approach

If national output is estimated either by using final product approach or value

addition approach it should provide the same value.

EX-

Firms	Product	Market price
1	Cotton	500
2	Threads	650
3	Fabric	900
4	clothes	1250
		Final Product Value

Inclusion of the value of goods and services to the national accounts which have passed the final stage of production process or goods and services that can be consumed or invested without processing further is called the final **product approach**.

According to the final product approach, the value of the total output is estimated by adjusting the value of change in inventory to the value of the sum of consumer goods and capital goods production.

Although the final product approach is used to eliminate the problem of multiple counting error it also has problems. This happens due to the difficulty of identifying consumer goods and intermediate goods separately.

Example:- Although sugar is considered as a final good it can be an intermediate good for a confectionary producer.

2. Value addition approach

The most suitable approach to estimate the value of national output avoiding these weaknesses of final product approach would be the value addition approach.

Gross value added (GVA) means the value received after deducting the value of intermediate inputs (Intermediate consumption) from the gross value of output (GVO).

EX-

Firms	Product	Market price	
1	Cotton	500	← 500
2	Threads	650	← 150
3	Fabric	900	← 250
4	clothes	1250	← 350
		1250	←

Final Product Value

Value addition approach

Analyses the related concepts and the estimation of gross domestic product following product approach

Gross domestic production

Gross Domestic Product means the value of all final goods and services produced within a particular period of time (Normally within one year), within the geographical boundaries of the country.

Gross National production

Gross National Product (GNP) is the total value of all finished goods and services produced by a country's citizens in a given financial year, irrespective of their location. GNP also measures the output generated by a country's businesses located domestically or abroad.

$$\text{GNP} = \text{GDP} + \text{Net primary income of rest of the world}$$

GNP vs. GDP



Value of all goods & services made by a country's residents & businesses, regardless of production location



Measures production inside of a country, no matter who makes it

How to calculate GDP by using Product approach

Net taxes

Value added in the agricultural sector	XXX
Value added in the industrial sector	XXX
Value added in the service sector	XXX
Gross Value Added	XXXX
Taxes on products (+)	XXX
Subsidies (-)	XXX
Gross Domestic Production (GDP)	

1- Do calculations based on the following economic data.

Value added in the agricultural sector	100
Value added in the industrial sector	250
Value added in the service sector	210
Taxes on products (+)	200
Subsidies (-)	100
Imports	400

1- Calculate Net taxes

2- Calculate Gross domestic production

2- Do calculations based on the following economic data.

Value added in the agricultural sector	150
Value added in the industrial sector	50
Value added in the service sector	350
Taxes on products (+)	50
Subsidies (-)	10
Exports	80

1- Calculate Net taxes

2- Calculate Gross domestic production

3- Do calculations based on the following economic data.

Value added in the agricultural sector	210
Value added in the industrial sector	390
Value added in the service sector	500
Taxes on products (+)	20
Subsidies (-)	10
Imports	100

1- Calculate Net taxes

Calculate Gross domestic production

4- Do calculations based on the following economic data.

Value added in the agricultural sector	355
Value added in the industrial sector	105
Value added in the service sector	400
Taxes on products (+)	200
Subsidies (-)	100
Net income	630

1- Calculate Net taxes

2- Calculate Gross domestic production

5- Do calculations based on the following economic data.

Value added in the agricultural sector	900
Value added in the industrial sector	800
Value added in the service sector	200
Taxes on products (+)	750
Subsidies (-)	250
Net income output	450

1- Calculate Net taxes

2- Calculate Gross domestic production

Any taxes on products are not included in basic price.

Basic price can be estimated as follows.

$$\text{Basic price} = \text{Producers' price} - \text{Net taxes on products}$$

(Net taxes on products = Taxes - Subsidies)

- Producers' price

The producers' price is the price receivable by the producer when releasing the product from the production place towards retailers.

It is the value received after deducting the value of subsidies provided over producers plus direct taxes on products at basic price.

Producers' price can be estimated as follows

$$\begin{array}{rcl}
 & \text{Basic prices} & \leftarrow \\
 & \boxed{+} & \\
 & \text{Taxes on products excluding invoiced VAT} & \\
 & - & \\
 & \text{Subsidies on products} & \\
 & = & \\
 & \boxed{\text{Producers' prices}} & \leftarrow \\
 & + & \\
 & \text{VAT not deductible by the purchaser} & \\
 & + & \\
 & \text{Separately invoiced transport charges} & \\
 & + & \\
 & \text{Wholesalers' and retailers' margins} & \\
 & = & \\
 & \boxed{\text{Purchasers' prices}} & \leftarrow
 \end{array}$$

The value of domestic income is obtained under the income approach by adding all factor earnings contributed towards production process

Factor incomes can be classified under the five main headings

• Employee compensation

Employee compensation means the earnings of supplying labour for the production process. It can be classified under three headings.

- ❖ Wages / salaries
- ❖ Payments for social protection systems by the employers
- ❖ Insurance facilities, health facilities, disaster allowances, labour compensation for employees

• Rental income

Rental income means the incomes gained mainly by renting real assets and natural resources.

Rental income consists of the following sources.

- ❖ Rental income earned by renting fixed assets and natural resources
- ❖ Rental income earned by renting houses and the assumed rent for the houses where owner lives
- ❖ Income for intellectual property

- Net interest income

Interest income means the payment made by the business sector for the suppliers of financial capital on behalf of the usage of financial capital.

- Corporate profits

The profits of the owners of business firms registered as companies are called corporate profits.

Profits and entrepreneurial income consists of the following sources.

- ❖ Corporate taxes
- ❖ Dividends for owners of the companies
- ❖ Retained profits
- ❖ Total profits included in the national accounts

- Self-employment income/ mixed income

The incomes of sole enterprises not registered as companies, joint ventures, cooperatives, farmers and incomes of professionals are called mixed income.

Following SNA 2008 components of Sri Lanka's gross domestic income is classified in a different.

1. Compensation of employees(CE)

Salaries and wages, contribution towards social protection provided by employers, direct funds given by employers in the situations of the sick, the unemployed and the retired are called compensation of employees.

2. Operational surplus(OS)

Operational surplus is the income generated by corporate sector. (It consists of interest, rent and gross profits)

- Gross operational surplus means the balance gained after deducting all costs and net production taxes from the value of a product.

Main components of gross operational surplus are as follows.

- Net operational surplus
- Fixed capital consumption

3. Mixed income

Mixed income means the income generated by non-corporations of households and joint ventures.

4. Other taxes on products

Other taxes on products means the taxes imposed over the provision of administration services or public property. (Imposing of rules and regulations for production process, providing of administration needs and facilities for production process)

INCOME APPROACH

1-Compensation of employees	XXX
2-Gross operational surplus	XXX
* Net operational surplus	XXX
* Fixed capital consumption	XXX
3-Mixed income	XXX
4- Other net taxes on products	XXX
G R O S S V A L U E A D D E D	XXXX
Taxes on product (+)	XXX
Subsidies (-)	XXX
G r o s s D o m e s t i c p r o d u c t i o n	XXXX

- 1- Gross operational surplus= Net operational surplus+ Fixed capital formation
- 2- Gross value added= Compensation of employees+ Gross operational surplus+ Mixed income+ Other net taxes on products
- 3- Net taxes= Taxes-subsidies

The combination Product approach and Income approach

Product approach

Income approach

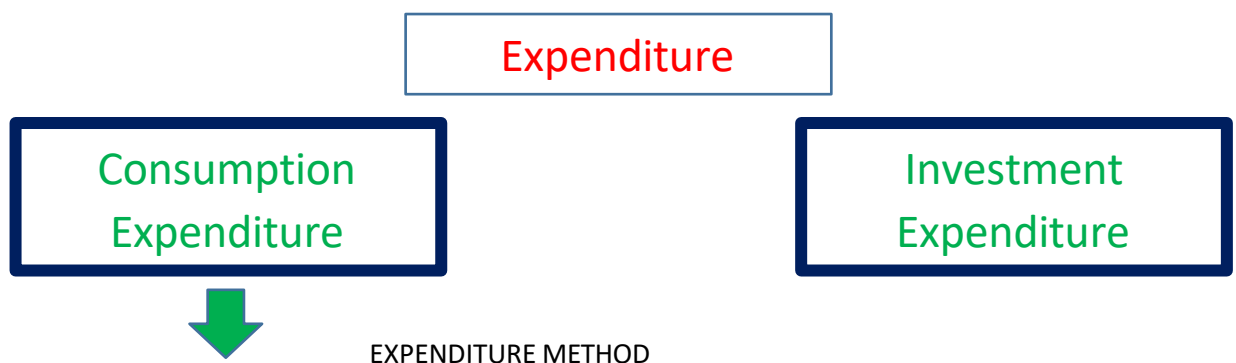
Value added in the agricultural sector	XXX	Compensation of employees	XXX
Value added in the industrial sector	XXX	Gross operational surplus	XXX
Value added in the service	XXX	Mixed income	XXX
	XXX	Other net taxes on products	XXX
Gross value added			
Taxes on products (+)			
subsidies (-)			
Gross domestic production			

EX 1

Compensation of employees	80
Net operational surplus	20
Fixed capital consumption	60
Mixed income	300
Other net taxes on products	100
Taxes on product	1000
Subsidies	950
Net primary income of rest of the world	-100
Value added in service sector and agricultural sector	20

- 1- Calculate net taxes
- 2- Gross operational surplus
- 3- Gross value added
- 4- Find the value added in industrial sector
- 5- Gross domestic production
- 6- Gross national product

Gross Domestic Product is calculated through the expenditure approach by estimating the value of the total amount spent to purchase the total output of an economy at a particular period of time.



1-Final consumption expenditure (FCE)	
* Household consumption expenditure	
* Government consumption expenditure	
* Services provided to households with non-profits	
2- Gross investment (Gross domestic capital formation)	
* Fixed capital formation	
* Change in inventory	
* Change in value	
Gross domestic expenditure at market price	
NET EXPORTS = Exports - Imports	
GROSS DOMESTIC PRODCUTION	

EX 1

Household consumption expenditure	200
Government consumption expenditure	100
Gross domestic capital formation	500
Exports	200
Imports	50

- 1- Final consumption expenditure (FCE)
- 2- Net Exports
- 3- Gross domestic expenditure at market price
- 4- Gross domestic product

EX 2

Household consumption expenditure	1450
Government consumption expenditure	2300
Gross domestic capital formation	550
Exports	2400
Imports	3000
Net primary income of the rest of the world	2000

- 1- Final consumption expenditure (FCE)
- 2- Net Exports
- 3- Gross domestic expenditure at market price
- 4- Gross domestic product
- 5- Gross national product

EX 3

Household consumption expenditure	500
Government consumption expenditure	700
Gross domestic capital formation	1500
Exports	5000
Imports	4900
Net primary income of the rest of the world	-200

- 1- Final consumption expenditure (FCE)
- 2- Net Exports
- 3- Gross domestic expenditure at market price
- 4- Gross domestic product
- 5- Gross national product

EX 4

Household consumption expenditure	350
Government consumption expenditure	700
Gross domestic capital formation	800
Exports	1000
Imports	1500
Net primary income of the rest of the world	600

- 1- Final consumption expenditure (FCE)
- 2- Net Exports
- 3- Gross domestic expenditure at market price
- 4- Gross domestic product
- 5- Gross national product

★ Gross National Income

The concept of gross national product is not being used today. The reason for this is when the value of net primary income of rest of the world is adjusted to the value of gross domestic product the value obtained is not being the value of the product but is the value of the income.

Gross National Income=Gross Domestic Product + Net primary income of rest of the world

Net primary income of the rest of the world= Primary income received from other countries- Primary income paid to other countries

EX 03

- 1- The Nominal gdp of an economy is 20000 and Net primary income of the rest of the world is 2500 calculate the Gross National Income.

- 2- The Nominal gdp of an economy is 54000 and Net primary income of the rest of the world is 1000 calculate the Gross National Income.

- 3- The Nominal gdp of an economy is 100000 and Net primary income of the rest of the world is 20000 calculate the Gross National Income.

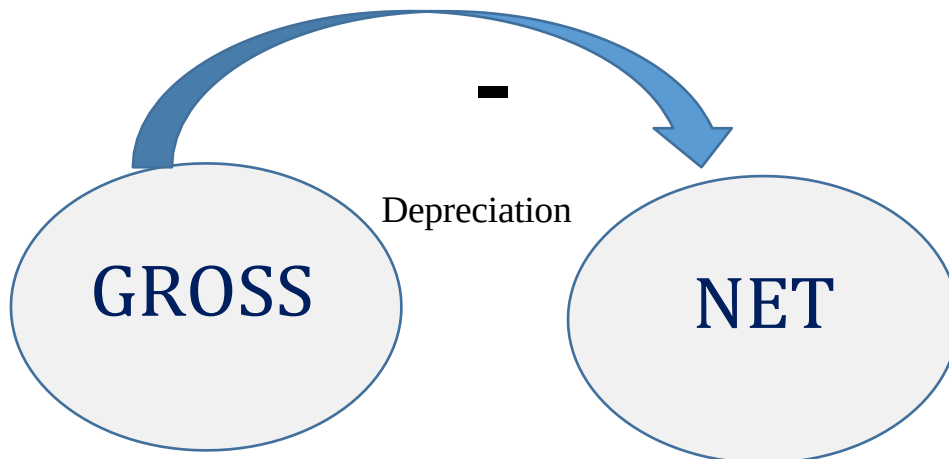
- 4- The Nominal gdp of an economy is 1800 and Net primary income of the rest of the world is -500 calculate the Gross National Income.

★ Net national income

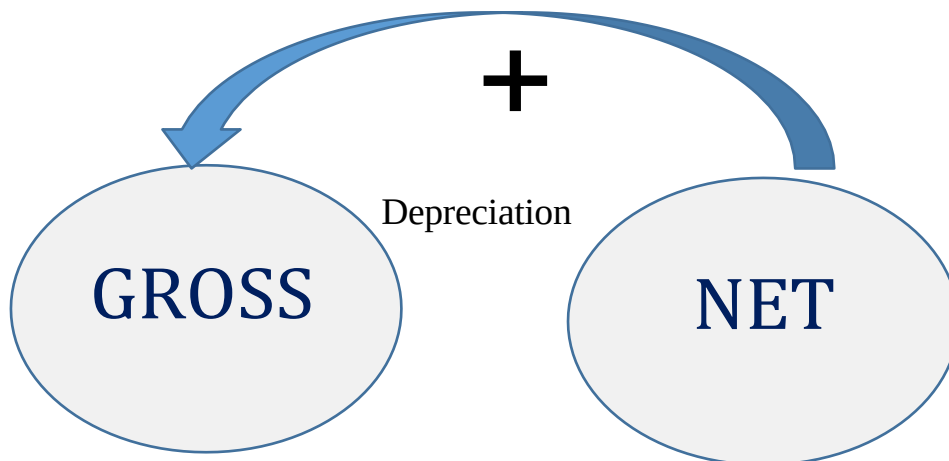
Net national income is the value obtained by deducting the fixed capital consumption from the value of gross national income.

Net National Income=Gross National Income - Fixed Capital Consumption
(Fixed Capital consumption= Capital depreciation)

Converting Gross concept into a Net concept



Converting Net concept into a Gross concept



- 1- The Nominal gdp of an economy is 25720 and Fixed Capital consumption 100 is Net domestic product.

- 2- The Nominal gdp of an economy is 2000 and Fixed Capital consumption 120 is Net domestic product.

- 3- The Nominal gdp of an economy is 3450 and Fixed Capital consumption 400 is Net domestic product.

- 4- The Nominal gdp of an economy is 4700 and Capital depreciation 100 is Net domestic product.

- 5- The Nominal gdp of an economy is 6000 and Capital depreciation 1000 is Net domestic product.

- 6- The gdp at market price of an economy is 85000 and Capital depreciation 2500 is Net domestic product.

- 7- The Gross domestic expenditures of an economy is 60000 and Capital depreciation 2000 is Net domestic Expenditures.

Importance of national income accounting
and limitations of national income accountin

★ Main uses of national accounting data is shown below.

- To assess the duty of an economy
- To measure economic growth
- To estimate per capita GDP
- To reveal structural characteristics of an economy
- To reveal functional income distribution
- For international comparisons
- To identify functional relationships between economic actors
- To reveal resource stock and resource utilization
- To obtain information for economic management
- To forecast behavior of macroeconomic variables

★ Limitations of National Accounting

○ Some of productive economic activities are not included in national accounts.

- Although goods production activities take place within a household included in national accounts the services rendered by households for their own consumption are not included in national accounts considering them as noneconomic activities. (Unpaid services performed within a house)
- Although hidden economic activities and illegal economic activities are included in national accounts the data relating to these activities could not be obtained due to absence of records, absence of information and evading form laws.
- Housewives work is not included in national accounts.
- Productive economic activities taken place within informal economy not included in national accounts.

- SoNon consideration of effect over environmental resources due to production.
- Non consideration of changes in the productivity of government services.
- Happiness gained through leisure is not included in national accounts.
- Interest payments by the government is not included in estimating national accounts.